

Disk Analyzer Technical Overview

Lightweight Edge Collector for Multi-Pod Ecosystems

A single-run utility designed for recursive file-system audits, metadata extraction, and multi-channel security reporting.

Filling the Visibility Gap

Edge Autonomy

Disk Analyzer operates at the pod level to capture raw storage metrics that centralized orchestrators often miss or abstract away.

Non-Intrusive Design

Engineered as a **low-impact task** rather than a persistent daemon, ensuring no interference with primary business logic.

Single-Run Lifecycle

Execution Flow

- Initialize Scan Roots via Environment
- Recursive OS Walk & Size Calculation
- Metadata Synthesis (Hostname/Time)
- Synchronous Reporting & Exit

Performance Specs

- Memory Footprint: < 50MB (Variable)
- Runtime: Optimized $O(n)$ Traversal
- Zero Persistent Dependencies

Recursive OS Traversal

The engine utilizes `os.scandir` for high-performance directory listing and `os.walk` for deep recursive calculation, featuring robust error handling for restricted paths.

```
# Traversal & Byte Conversion Logic
def dir_size_bytes(path):
    total = 0
    for current_root, dirs, files in os.walk(path):
        total += sum(os.stat(f).st_size for f in files)
    return bytes_to_gb(total)
```

Telemetry Ingestion

Backend Integration

Reports are serialized into JSON and delivered to the Multi-Pod Backend via HTTP POST. Features configurable timeouts for unreliable networks.

Schema Validation

Payloads strictly adhere to the Pydantic model defined in the core backend, ensuring seamless compatibility across the ecosystem.

```
Endpoint: /report  
Payload: { hostname, timestamp, items }
```

Redundant SMTP Reporting

Daily Summaries

Independent of the API, the agent generates **EmailMessage** summaries. This provides a direct notification path for SOC managers.

- Top-N Heavy Directory Identification
- Threshold-Based Critical Markers

Security Fallback

Utilizing `smtplib` with TLS encryption, the daily report acts as a secondary audit trail if the dashboard telemetry is interrupted.

```
Marker: "HIGH USAGE" if > BIG_THRESHOLD
```

Environment-Aware Config

CLI Parameters

- `--dry-run`: Validate without delivery
- `--top-n`: Customizable summary depth
- `--big-threshold-gb`: Anomaly definition

Docker Readiness

Configured for containerization through ENV vars:
SCAN_ROOTS, EMAIL_USER, and
MULTIPOD_BACKEND_URL.

Deterministic Validation

Error Handling

Path access is protected by try-except blocks, preventing agent crashes when encountering `PermissionError` or broken symbolic links.

Integrity Checks

The Dry-Run mode generates full report summaries and email body previews in standard output for local validation before production deployment.

Expansion Roadmap

Phase I: Enhanced Metadata

- File Entropy Analysis for Encryption Detection
- Owner/Permission Change Monitoring
- Integration of eBPF for I/O Monitoring

Phase II: Advanced Protocol

- gRPC Stream Support for Large Environments
- JWT-Signed Payloads for Anti-Spoofing
- Local Differential Privacy for File Metadata